

Superstore Sales SQL Analysis

End-to-End Retail Performance, Profitability & Root-Cause Investigation

PostgreSQL

Sales Analytics

Business Intelligence

9,994 Records

1. EXECUTIVE PROJECT OVERVIEW

This project conducts a comprehensive end-to-end data analysis using SQL (PostgreSQL) on the Superstore retail dataset. The analysis spans basic dataset exploration, regional revenue mapping, customer segmentation, discount sensitivity, and controlled root-cause analysis to pinpoint unprofitable business drivers.

TOTAL RECORDS

9,994

2011 – 2014

TOTAL SALES

\$2.30M

Across 4 Regions

UNIQUE PRODUCTS

1,841

17 Sub-Categories

GEOGRAPHIC REACH

531

Cities Covered

2. DATASET QUALITY & STRUCTURE (01_EXPLORATION.SQL)

Data integrity was verified prior to downstream aggregations. The postal code, city, sales, and profit fields were checked for potential missing or null values.

```
-- Data Quality Check: Verify no missing values in key metrics
SELECT COUNT(*) - COUNT(postal_code) AS null_postal_codes,
COUNT(*) - COUNT(city) AS null_cities,
COUNT(*) - COUNT(sales) AS null_sales,
COUNT(*) - COUNT(profit) AS null_profits
FROM superstore;
-- Output: 0 NULLs across all inspected columns
```

Project Repository Structure

File Name	Primary Focus	SQL Concepts Employed
01_exploration.sql	Dataset size, coverage, product hierarchy & quality checks	COUNT, DISTINCT, MIN, MAX
02_sales_analysis.sql	Geographic performance, category rank, YoY growth & trends	GROUP BY, JOIN, RANK(), EXTRACT()
03_profit_analysis.sql	Profitability drivers, loss detection & controlled tests	HAVING, Window Functions, Controlled Filtering

Key Finding: Data Quality

All transaction records from 2011 through 2014 are structurally complete with zero null values in primary numeric and geographic fields, enabling reliable multi-dimensional analysis.

3. GEOGRAPHIC SALES PERFORMANCE

Revenue distribution was analyzed across regions and state levels using window functions to isolate top-performing territories.

```
-- Top-selling state per region using RANK() WITH state_sales AS ( SELECT region,
state, ROUND(SUM(sales), 2) AS total_sales, RANK() OVER (PARTITION BY region ORDER BY
SUM(sales) DESC) AS rnk FROM superstore GROUP BY region, state ) SELECT region, state,
total_sales FROM state_sales WHERE rnk = 1 ORDER BY total_sales DESC;
```

Top Performing State by Region

Region	Top State	Total Regional Sales	Top State Revenue
West	California	\$725,457.93	\$457,687.68
East	New York	\$678,781.24	\$310,876.20
Central	Texas	\$501,239.89	\$170,187.98
South	Florida	\$391,721.90	\$89,473.73

4. CATEGORY & SUB-CATEGORY SALES BREAKDOWN

Technology leads total revenue across the three primary categories, supported strongly by Phones and Chairs at the sub-category level.

Category Performance

Category	Sales (\$)
Technology	\$836,154.10
Furniture	\$741,999.98
Office Supplies	\$719,046.99

Top 5 Sub-Categories

Sub-Category	Sales (\$)
Phones	\$330,007.10
Chairs	\$328,449.13
Storage	\$223,843.59
Tables	\$206,965.68
Binders	\$203,412.77

5. TOP REVENUE-GENERATING CUSTOMERS

Customer Name	Primary Segment	Total Revenue (\$)
Sean Miller	Consumer	\$25,043.07
Tamara Chand	Corporate	\$19,052.22
Raymond Buch	Consumer	\$15,117.35
Tom Ashbrook	Home Office	\$14,595.62
Adrian Barton	Consumer	\$14,473.57

6. SALES TRENDS & SEASONALITY ANALYSIS

Aggregating order date timestamps reveals distinct Q4 spikes driven by holiday purchasing cycles, with November and December delivering peak transaction volumes.

Monthly Sales Peak Ranking

Rank	Month	Total Revenue (\$)
1	November	\$349,120.08
2	December	\$332,177.20
3	September	\$309,770.12
4	March	\$199,253.01
5	October	\$197,115.24

Seasonal Revenue Dynamics

Sales consistently climb from Q1 through Q4. The last four months of the year generate over 40% of total annual sales volume.

Q1 represents a recurring post-holiday dip, with February averaging lowest overall volume (\$60,172.66).

7. YEAR-OVER-YEAR (YOY) GROWTH CALCULATIONS

Using self-joins, annual revenue was calculated alongside year-over-year performance metrics to measure growth momentum.

```
-- YoY Growth Query via Self-JOIN
SELECT curr.year, curr.total_sales AS current_sales,
       prev.total_sales AS prev_sales, ROUND(((curr.total_sales - prev.total_sales) /
       prev.total_sales) * 100, 2) AS yoy_growth
FROM ( SELECT EXTRACT(YEAR FROM order_date) AS year,
      SUM(sales) AS total_sales
      FROM superstore
      GROUP BY year ) curr
LEFT JOIN ( SELECT EXTRACT(YEAR FROM order_date) AS year,
      SUM(sales) AS total_sales
      FROM superstore
      GROUP BY year ) prev
ON curr.year = prev.year + 1
ORDER BY curr.year;
```

Annual Growth Summary

Year	Annual Revenue (\$)	Prior Year Revenue (\$)	YoY Growth (%)
2011	\$484,247.56	—	—
2012	\$470,532.46	\$484,247.56	-2.83%
2013	\$608,474.08	\$470,532.46	+29.32%
2014	\$733,946.97	\$608,474.08	+20.62%

Trajectory Strategic Note

After a slight pullback in 2012 (-2.83%), the store achieved significant acceleration, expanding total annual turnover by +51.5% from 2011 (\$484K) to 2014 (\$733K).

8. PROFITABILITY & LOSS-MAKING ANALYSIS

High sales volume does not guarantee profitability. While Technology generated \$145,454.95 in net profit, specific products and sub-categories incurred substantial losses.

Top Loss-Making Products

Product Name	Category	Sales (\$)	Profit (\$)
Cubify CubeX 3D Printer Double Head Print	Technology	\$11,099.96	-\$8,879.97
Lexmark MX611dhe Monochrome Laser Printer	Technology	\$16,829.90	-\$4,589.97
Cubify CubeX 3D Printer Triple Head Print	Technology	\$7,999.98	-\$3,839.99
Chromcraft Bull-Nose Wood Oval Conference Table	Furniture	\$9,917.64	-\$2,876.11

Sub-Category Deficit Alert

The **Tables** sub-category recorded an aggregate net deficit of **-\$17,725.59**, making it the most unprofitable line item in the entire retail inventory.

9. DISCOUNT IMPACT ANALYSIS

Evaluating profit levels against discount percentages reveals a clear tipping point where discounting destroys business margins.

Average Profit by Discount Level

Discount Rate	Avg Profit (\$)	Margin Status
0%	\$66.90	Profitable
10%	\$96.06	Optimal
15%	\$27.29	Reduced
20%	\$24.70	Reduced
30%	-\$45.68	Loss Making
32%	-\$88.56	Loss Making
40%+	-\$111.93	Severe Loss

```
-- Discount Profitability SQL
SELECT discount, ROUND(AVG(profit), 2) AS
avg_profit FROM superstore GROUP BY
discount ORDER BY discount;
```

Key Insight: Any discount rate of 30% or higher results in negative net returns on average.

10. CUSTOMER SEGMENT ANALYSIS

Performance varies substantially between customer types. Home Office accounts for lower total volume but yields the highest relative margin.

Customer Segment	Customers	Sales (\$)	Profit (\$)	Profit Margin (%)
Consumer	409	\$1,161,401.34	\$134,119.33	11.55%
Corporate	236	\$706,146.44	\$91,979.45	13.03%
Home Office	148	\$429,653.29	\$60,299.01	14.03%

11. SHIPPING MODE & CONTROLLED VARIABLE ANALYSIS

First Class shipping yields higher raw margins (13.93%). To verify whether shipping mode directly influences margins or if category mix skews the metric, a controlled experiment was run holding Category and Discount constant.

```
-- Controlled Analysis: Technology category with 0% discount
SELECT ship_mode,
category, discount, COUNT(*) AS total_orders, ROUND(AVG(sales), 2) AS avg_order_value,
ROUND(SUM(profit)/SUM(sales)*100, 2) AS profit_margin_pct FROM superstore WHERE category =
'Technology' AND discount = 0 GROUP BY ship_mode, category, discount ORDER BY
profit_margin_pct DESC;
```

Controlled Margin Results (Technology @ 0% Discount)

Shipping Mode	Orders	Avg Order Value (\$)	Controlled Margin (%)
First Class	141	\$418.53	36.66%
Same Day	41	\$620.60	35.61%
Standard Class	488	\$494.29	34.08%
Second Class	163	\$393.02	30.34%

12. KEY TAKEAWAYS & STRATEGIC ACTION ITEMS

- Implement Discount Caps:** Cap promotional discounts at 20%. Discounts ≥30% erase profitability across all sub-categories.
- Address Deficit Sub-Categories:** Re-evaluate vendor pricing and freight costs on Tables (-\$17.7K loss) and high-end 3D Printers.
- Target High-Margin Segments:** Increase marketing focus on the Home Office segment (14.03% profit margin).
- Methodological Lesson:** Controlled SQL comparisons prevent misleading conclusions caused by confounding multi-variable aggregations.